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Introduction

We are not yet ready—if we ever will be—for a grand theory of the survey process that explains all aspects of question answering with a limited number of powerful theoretical perspectives. In this book we present an eclectic theoretical framework drawn from psycholinguistics and social and cognitive psychology, and we review the growing literature on questionnaire design that uses this framework to study response errors in surveys.

We start with a dual conception of the survey. On the one hand, a survey is a social encounter. On the other hand, it is a series of cognitive tasks to be performed by respondents. When we call it a social encounter, we mean that interviewers conduct interviews in a form that is very similar to that of ordinary conversations and, as such, partakes of the linguistic and social rules that govern such conversations. To be sure, a survey is a special type of conversation with characteristics that set it apart from many other conversations, but it is a conversation nonetheless and needs to be understood as a conversation. A survey is also a voluntary social encounter between strangers and is subject to the rules that govern social relations between strangers. Thus, understanding the rules that govern conversations and social encounters in general should help us understand how survey questions are being understood and answered.

The vehicle through which survey questionnaires are delivered is language. Understanding how people comprehend speech and written material deepens our understanding of the ways in which questionnaire design affects people's answers. Language comprehension is based not only on formal structures like syntax, but more importantly, on pragmatic factors that may deeply affect meaning. A study of the effects of question wording has to focus on the factors that influence respondents' comprehension and on how the inferred meaning may differ from the researcher's intended meaning. Variations in contexts may result in different respondents answering, in effect, different questions (in such cases, the answers cannot be compared) or even questions that the researcher never intended to ask.

Trying to understand how respondents comprehend survey questions leads inevitably to a more basic search for information about how people understand the world around them and how they communicate with one another. Cognitive psychology has a rich research literature on memory and information processing, processes that are fundamental to understanding how respondents answer survey questions. It may seem as if a long distance separates the simple phenomenon of asking survey questions and the complex cognitive processes that individuals go through in answering them. But we maintain that only when the social and cognitive complexities of the response process are understood will we have a good understanding of the quality of survey data and know when response effects adversely affect the results of our surveys and when they can safely be ignored.

Outline of the Book

This book begins with an exploration of the methodological approaches researchers use to understand question characteristics, the cognitive processes, and respondents' difficulties in answering questions. Because these methods can be applied independently of the particular question answering process used and because they are

very useful as supplements to traditional pretesting strategies, we review them in Chapter Two. Such methods as protocol analysis or cognitive interviews, in which respondents "think aloud" as they answer survey questions, give insight into the ways in which they understand questions, search their memories for relevant information, form judgments, and edit their answers. The limitations of these methods are also discussed. This chapter is likely to be especially useful to survey research practitioners.

Chapter Three presents an outline of cognitive theorizing, which underlies the empirical work reviewed in later chapters. Several very similar models of the survey process have been proposed. In the chapter, the models are described and their implications for survey practice explored. The basic concepts introduced are elaborated on in more detail as our discussion proceeds to more specific issues.

The subsequent three chapters deal with the context effects in question asking from a cognitive perspective. Chapter Four reviews different sources of context effects and describes their influence on the cognitive tasks involved in the question answering process. As many researchers have noted, however, context effects often seem inconsistent. Sometimes they move responses in the direction of previous information (usually referred to as "assimilation effects") and sometimes away from previous information (usually referred to as "contrast effects"). Chapter Five reviews in more detail why context effects may turn out to be assimilation effects or contrast effects and offers guidelines to help the researcher. Information that may affect responses is contained not only in previous questions and information but also in the response alternatives offered. Chapter Six examines context effects that arise from response alternatives and the order in which they are presented. Cognitive theory is used to explain why effects are sometimes based on the first items presented (primacy effects) and at other times on the last few items presented (recency effects).

The next four chapters relate to reports of behavior, or what is currently called "autobiographical memory." Chapter Seven deals

with the storage and retrieval of autobiographical memories. In it we review some theories about how information is stored and how the organization of memory can affect accurate recall and then discuss some experiments that support these theories.

Many survey questions about behavior refer to events that occurred within a specified time period, such as, "How many times have you been to the doctor in the last three months?" Special cognitive effort is required in remembering not only that an event occurred but also when it occurred. Chapter Eight treats theories of memory for time and examines the reasons for systematic biases in reports on when events occurred.

When the memory task set by the survey questions is too difficult, respondents turn to estimation. In Chapter Nine we discuss when and how estimation occurs and some of the strategies respondents use to make estimates. We also discuss the ways in which behavior regularity may help or hinder retrieval of information about that behavior.

Reports on behavior come not only from respondents themselves but also from others who live in the same household or know them well. The task of reporting about others within a household or an organization adds still new cognitive complexities. In Chapter Ten we discuss these issues and present experiments describing comparisons of cognitive processes in self-reporting and proxy reporting.

Finally, in Chapter Eleven we summarize what we think we have learned from the application of cognitive theory to the survey process and review our social-cognitive framework of survey interviewing.

A Brief Research History

Before proceeding with the book, it is useful to set the stage with a short review of the research on survey questionnaire design. Since the earliest days of survey research, investigators have worried about the possible biasing effects of the data-collection process. The inter-

viewers, the wording of questions, the order in which questions are asked—all have been seen as possible sources of measurement error. Indeed, it is probably fair to say that as survey research began in the 1930s and 1940s, more attention was given to these sources of error than to sampling error. This early work laid the foundations for some of the research described in this book.

Even with the relatively crude sampling methods of the time—quotas for respondents with different characteristics such as gender, age, and so on—the success of polls using these methods in the 1936 presidential race between Franklin D. Roosevelt and Alfred Landon established the superiority of scientific sampling over the self-selected methods typical of the media polls at that time. In 1936, the *Literary Digest*, using a huge but badly biased sample of ten million predicted a Republican victory. Yet the Democrats went on to win all but two states. At the same time, the new survey organizations such as Gallup, Roper, and Crossley, with a more "scientific" sample of only fifteen hundred, predicted the election's outcome correctly. Possible biases arising from different sampling methods, although discussed and known in the statistical literature, were not the subject of much research among the small group concerned with survey research as it grew during the late 1930s and early 1940s.

The failure of the polls to predict the outcome of the 1948 election—in which Harry Truman defeated Thomas Dewey—focused the attention of the academic community, as well as that of the media and the polling industry itself, on methodological problems. Attention to sampling methods was particularly sharp, and a lively debate about the relative merits of probability and quota sampling dominated the discussion. Interviewer bias, intertwined with the problems of quota sampling, was a source of great concern even before the 1948 debacle but gained renewed interest after it.

The Social Science Research Council (SSRC) sponsored several large projects to examine the methodology of surveys, particularly as practiced in the political arena. These efforts culminated in two important books. The first was a study done immediately after

the 1948 election that looked at what went wrong with the polls in that election. While reviewing the most likely steps in surveys where errors may occur, the authors concluded that there were two primary causes of errors: sampling and interviewing methods and forecasting methods (Mosteller, Hyman, McCarthy, Marks, & Truman, 1949).

Because most early surveys were concerned with attitudes and opinions and such “subjective” topics were thought to be particularly susceptible to influence by interviewers, much of the methodological concern in the late 1940s was about possible interviewer effects. Did the interviewers’ personal opinions influence the respondents? Were the interviewers consciously or unconsciously behaving in ways that biased the results? Did the personal characteristics of the interviewers make a difference? These questions lay at the heart of the second of the two SSRC projects and resulted in Hyman’s classic work, *Interviewing in Social Research* (1954). The results of this project, which was based on a large number of individual studies at the National Opinion Research Center, laid to rest the worst fears of those who believed that opinion research was subject to rampant bias introduced by interviewers and effectively exhausted for some time any in-depth research on interviewer effects.

During the 1960s, with the rapid growth of a variety of governmental transfer and training programs designed as part of the War on Poverty campaign, demand for evaluation of these programs grew. A new industry of evaluation research that used the method of sample surveys to collect the necessary data was created. In these surveys, the focus shifted away from opinions and toward behaviors. Survey methods were used to chart the experiences of people enrolled in both demonstration and operational programs, and sophisticated social experiments were conducted to try out different policy ideas such as health insurance plans, housing allowances, early childhood education programs (notably Head Start), and income maintenance programs. The data’s accuracy became of vital concern because large sums of money would be distributed on that

basis. Thus methodological interest shifted toward sources of measurement error when measuring behavior rather than attitudes or opinions.

Studying Measuring Instruments

As a result of intensive methodological work, survey sampling developed into a scientifically based enterprise with clearly specified theoretical foundations and codified rules of application (Kish, 1965; Sudman, 1976). To a lesser extent, this was also true for survey interviewing. Although the theoretical basis of interviewing is still less rigorously developed than the theoretical basis of sampling, considerable agreement has been reached on the proper conduct of survey interviews and standardized training programs have been developed (Cannell and Kahn, 1968; Weinberg, 1983). In contrast, question wording and questionnaire design have largely remained an “art,” governed by in-house traditions acquired by personal experience. Although many researchers explored the impact of question wording and question order on survey responses (for early reviews see the seminal volumes of Cantril, 1944, and Payne, 1951), the work conducted in this domain suffered from a lack of theoretical perspective. Articles appeared that demonstrated particular effects of particular question wordings and question orders, but no consistent body of findings or rules alerted the investigator to the types of situations in which wording or order effects were likely to appear. This resulted in a provocative collection of heterogeneous response effects yet their implications beyond the specific question under study remained uncertain. In the absence of a theoretical framework, it seemed that research on question wording and related factors had to start over with each new question asked.

Not surprisingly, many researchers lost interest in the issue and accepted that question wording would affect the distribution of the responses and that little could be done about it. Moreover, researchers hoped that question characteristics would primarily affect univariate distributions and not the observed relationship between

variables. This hypothesis became known as the *form-resistant correlations assumption* (Stouffer & De Vinney, 1949). Accordingly, respondent characteristics and interviewer behavior seemed to provide more promising research arenas, whereas interest in questionnaire variables declined in the mid-1950s, not to be revitalized until the 1970s.

Several developments contributed to the revitalization. First, the assumption of form-resistant correlations did not hold up. In a seminal paper, Schuman and Duncan (1974) demonstrated that the relationship between variables *does* depend on the specific question asked, thus bringing question form back into focus. Second, increasing interest in tracking social change and determining the impact of social programs required interpretable longitudinal data, giving rise to concern about the comparability of answers given at different points in time to the same question in different questionnaires. This concern was compounded by the fact that univariate distributions were of crucial importance for many of the substantive issues addressed in evaluation studies.

In this context, in the early 1970s two of the authors of the present work (Bradburn and Sudman) began work on an extensive meta-analysis of the response effects literature (Sudman & Bradburn, 1974). In order to organize that review, they proposed a simple model for understanding the sources of response effects. The model started with the view that the survey interview is a special kind of conversation between individuals who speak the same language. It is a "conversation with a purpose," as Bingham and Moore (1934) described it, because it has a definite structure and particular rules that differentiate it from other conversational forms such as casual conversation between friends or instrumental conversation between colleagues, for example.

The interview structure consists of a social situation in which one person (the interviewer) has the role of asking the questions, a second person (the respondent) has the role of answering the questions, and a third person (the researcher) defines the task. The researcher has a particular purpose in mind, that is, to obtain infor-

mation to answer some general practical or scientific questions. Thus the interview is in some ways like all conversations, in some ways like many conversations, and in some ways like no other conversation.

Methodological research on the survey process has tended to focus on aspects of the process that are unique to a particular survey or that resemble other surveys but ignore aspects of the process that are shared with other types of conversations. As we shall see in Chapter Three, the authors of the present work believe that consideration of the ways in which survey interviews adhere to the rules of general conversation helps us understand some response effects.

By conceptualizing the survey interview as a social system with two roles united by a common task, Bradburn and Sudman were able to identify three primary sources of response effects—those stemming from the interviewer, those coming from the respondent, and those coming from the task itself, particularly the questionnaire and the context within which it is perceived. The meta-analysis suggested that the largest effects were associated with the task itself, that is, the question asking and answering process. How powerful the impact of such task variables as question wording or question order could be was further illustrated by a series of split-ballot experiments on attitude questions by Howard Schuman and his collaborators, subsequently reviewed in Schuman and Presser's (1981) *Questions and Answers in Attitude Surveys*.

Together, these lines of work indicated that the most important part of the process is the wording of the question. The wording of the question is what the respondent must process in order to understand what information is being sought or, for that matter, that it is *information* that is being sought and not, say, an expression of affect or some other type of communication, as might be the case in another conversation.

This point is so fundamental that its importance is often overlooked. To take a vivid but trivial example, the questions "How old are you?" and "When were you born?" have only one word in

common but are recognizably about the same topic. Estimates of the age distribution of the population based on responses to the two questions from a sample of that population will differ but not dramatically. However, if the two questions were "When were you born?" and "Where were you born?" the responses would be so different that the answers could not be compared even though these questions share more of the same words than do the two about age. The point is that the question's meaning is what is important. As Bateson (1984) pointed out, "A survey trades in meanings, and meanings are embodied in language. A survey consists of a transfer of meanings between the three participants through the medium of language."

Indeed, small changes in wording can alter the meaning fundamentally while extensive changes in wording may alter it only slightly. An early experiment by Rugg (1941) showed that small changes in wording, even in questions about the same topic, could produce dramatically different effects. He asked matched samples of respondents one of two questions: "Do you think the United States should allow public speeches against democracy?" or "Do you think the United States should forbid speeches against democracy?" When the question was one of "allowing" public speeches, 21 percent of respondents supported free speech. But when the question was one of "forbidding," 39 percent supported free speech. Rugg had no explanation for these findings except to note that "forbid" and "allow" were not exact opposites when used in these sentences even though they looked superficially as if they were. (For a cognitive explanation of this finding, see Hippler & Schwarz, 1986.) An adequate theory of response effects must be able to account for such differences.

Interviews entail much more than just questions, of course, including nonsemantic information. For example, there is the social context in which the interview takes place. It may take place in the home of a respondent who is selected as a member of a sample of the population. Or in the workplace with a respondent who is an employee of an organization about which the questions are asked.

Or in a researcher's office or laboratory with the respondent a "subject" in an experiment that is part of a study whose purpose is perceived to contribute to scientific knowledge.

In addition, social characteristics of the interviewer and the respondent, such as race or gender, may carry over into attitudes and affect relations in ways that are independent of the interview and reflect larger social beliefs. The mode of administration of the questions may also have important consequences on the responses: Is the interview conducted in person, on the telephone, or through self-administered questionnaires?

Many of these variables will have different effects depending on the interaction between their characteristics and the questionnaire's content. And, of course, there will be individual differences in the degree to which respondents are influenced by any of these factors. Nevertheless, task variables have the greatest impact on the responses obtained in survey interviews. In many ways, this is a fortunate state of affairs because it is the nature of the tasks presented to respondents that the researcher has the most control over.

The Application of Cognitive Theory to Questionnaire Construction

Focusing on the nature of the question asking and answering process naturally leads one to consider the cognitive processes involved in answering survey questions. Thus, interest turned to a more detailed conceptualization at the individual psychological level of what was involved in comprehending and answering survey questions, and by the late 1970s psychology was prepared to provide some reasonable information on these issues.

Having been dominated by the paradigm of behaviorism for nearly four decades, experimental psychology underwent a "cognitive revolution" in the 1970s, bringing mental processes back to center stage. In contrast to behaviorism's emphasis on stimulus-response sequences and the importance of rewards, the emerging paradigm of information processing emphasized complex mental

processes. Experimental psychologists turned from animal learning, mazes, and food-dispensing levers to topics like information acquisition, storage, and retrieval, and the technology of reaction time measurement. (See Hunt, 1993, for a popular history of this development, and Lachman, Lachman, & Butterfield, 1979, for a more rigorous discussion.) Building on earlier work in nonbehaviorist traditions, this paradigm shift quickly resulted in the development of testable models of memory, language comprehension, inferential reasoning, and judgment, among others (see Lachman, Lachman & Butterfield, 1979). Although these topics are of key importance to an understanding of the question answering process, these developments went largely unnoticed by the survey research community. As work within the information processing paradigm progressed, however, some psychologists began to wonder how what they learned in the laboratory applied to human behavior in natural settings, giving rise to more naturalistic studies, in particular in the domain of memory. (For examples see the contributions in Neisser & Winograd, 1988.) This increasing interest in "everyday cognition" rendered applied issues a legitimate arena for cognitive research, setting the stage for cognitive psychologists' collaborations with other disciplines.

The collaboration between cognitively oriented psychologists and survey researchers can be traced to the late 1970s. One of the earliest instances, if not the earliest one, was a seminar held in 1978 by the British Social Science Research Council and the Royal Statistical Society on problems in the collection and interpretation of recall data in social surveys (see Moss & Goldstein, 1979).

The next two important events occurred in the United States in 1980. The first was a workshop convened by the Bureau of Social Science Research in connection with its work in the redesign of the National Crime Victimization Survey sponsored by the National Institute of Justice and the Census Bureau (see Biderman, 1980). This workshop brought together cognitive scientists, survey statisticians, and methodologists to discuss what contributions cognitive scientists could make to understanding response errors in behav-

ioral reports. The second event was the establishment of a panel on the measurement of subjective phenomena by the Committee on National Statistics at the National Research Council/National Academy of Sciences. This panel produced two large volumes (Turner & Martin, 1982) that have stimulated considerable research on response effects involved in the measurement of subjective phenomena. This work complements the work that had been stimulated by the earlier seminars on measuring behavior or more "objective" phenomena.

Further impetus for the development of more cognitive approaches to the study of response effects came from a six-day seminar, "Cognitive Aspects of Survey Methodology" (Jabine, Straf, Tanur, & Tourangeau, 1984) in St. Michael's, Maryland, sponsored by the Committee on National Statistics and funded by the National Science Foundation.

In Europe, independent work was going on in parallel directions. Schwarz and his colleagues at the University of Heidelberg and ZUMA, a German social science research center, were beginning to carry out cognitive experiments related to questionnaires. A seminar involving European psychologists with some participation by American researchers took place at ZUMA in the summer of 1984 (see Hippler, Schwarz, & Sudman, 1987). Several cognitively based conceptions of the interview task came out of these conferences (for example, Strack & Martin, 1987; Tourangeau, 1984, 1987) and provided a basic conceptual framework for much of the subsequent empirical work.

In the decade since these initial conferences, the development of the field has been rapid and vigorous. Several U.S. Government statistical agencies including the Census Bureau, the Bureau of Labor Statistics, and the National Center for Health Statistics now have cognitive laboratories for questionnaire improvement. Schwarz and Sudman organized additional conferences: "Context Effects in Social and Psychological Research" in 1989, "Autobiographical Memory and the Validity of Retrospective Reports" in 1991, and "Methods for Determining Cognitive Processes Used to

Answer Questions" in 1993 (see Schwarz & Sudman, 1992, 1994, 1995). The Social Science Research Council organized the Committee on Cognition and Survey research, which sponsored a series of cognitive conferences (Tanur, 1992).

By now, several major survey centers, in the U.S. as well as in Europe, have established cognitive laboratories to help with questionnaire development, and the first degree-granting program in survey methodology (the University of Maryland–University of Michigan Joint Program in Survey Methods) offers courses in cognitive and social psychology as part of its curriculum.

Conclusion

The utility of applying cognitive psychology to the study of survey measurement error is now well established. The time seems right, therefore, to pull together some of the work that has been done and try to develop a theoretical basis for integrating the work to date. By so doing, we hope to set the stage for the kind of cumulative knowledge that is the hallmark of science.

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Methods for Determining Cognitive Processes and Questionnaire Problems

This chapter describes the methods developed to determine the cognitive processes involved when respondents answer survey questions and the difficulties caused by the research instrument. Schwarz and Sudman's (1995) *Answering Questions* describes these and other methods in more detail. There are other ways in which the chapter could be organized (Forsyth & Lessler, 1991). The organization we use sets the stage for the framework we develop more fully in the next chapter. We discuss in separate sections the methods used by researchers to learn how respondents interpret a question, retrieve information, form judgments, and edit their answers.

For each of these cognitive tasks, we describe the procedures used and evaluate their quality. Some researchers (Nisbett & Ross, 1980; Nisbett & Wilson, 1977) have challenged the idea that individuals can provide accurate introspective reports about their cognitive processes and have presented examples of experiments where respondents' reports about their thought processes were misleading. We will discuss these criticisms but believe the evidence is very strong that the methods that have been developed to date provide useful, although not perfect, information for understanding respondents' cognitive processes and improving questionnaires. These methods generally do not rely on respondents' introspective insights into their cognitive processes but elicit material from which